

Formalizing LCR, Unifying Standard Models

Unified Master Publication Manuscript

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Working Title

Formalizing LCR, Unifying Standard Models: A Receipt-Bound, Analog-Accessible, CAM/Crystal Publication Framework For Cross-Domain Physical Formalization

Abstract

This manuscript presents the FLCR series as one integrated scientific publication program. The work begins with a finite LCR carrier, correction surface, receipt discipline, and claim-lane governance; extends through lattice closure, CAM/crystal memory, forge actionization, TarPit mass/bond metrics, recursive Damascus folding, and Monster-prime boundary accounting; and culminates in Standard Model translation and grand-synthesis papers. The central methodological claim is that maximal scientific hypotheses can be stated explicitly while their proved parts, adapters, calibrations, and falsifiers remain separately tracked. The analog toolkit is not merely exposition: it is the access layer intended to make the same STEM structures reconstructable without reliance on electricity, internet, computers, or paper. Guardrails are therefore interpreted as automation controls for agents and publication systems, not as restrictions on human hypothesis formation. Open obligations are treated as channels between the present proof state and the larger claim, declaring the prestate from which subsequent exploration proceeds. The later papers are directed toward literal physics through receipt-bound adapters, accepted standard formalisms, calibration data, and falsification protocols.

Keywords

LCR; CAM; crystal projector; TarPit; claim lanes; analog STEM access; Standard Model translation; Monster; receipt-bound proof; open obligations; literal physics.

1. Introduction

The FLCR corpus is being assembled as a single publication-grade scientific argument rather than a loose collection of drafts. Each individual paper remains a modular source unit, but this master manuscript is the coherent peer-review surface: it states the full thesis, organizes the methods, routes evidence, displays results, and names the remaining obligations required for stronger physical claims.

The publication posture is deliberately maximal. The manuscript states the grand version of the intended claims, then displays what is currently proved, sourced, computed, calibrated, or falsifiable. It does not treat an obligation as a retreat from the claim. It treats an obligation as the channel connecting the present prestate to the next required derivation.

2. Source Material And Editorial Authority

All current artifacts are source material, but not all artifacts carry the same authority. The master manuscript uses the following source policy.

Source class	Content	Use rule
Full source	Canonical FLCR formal/companion/workbook files; SMD/SMB defining papers; master routing doctrine; receipt markdown/json generated in the current pass.	May be quoted, summarized, rewritten, or promoted according to claim lane.
Partial source	Legacy 00-32 papers, 1-80 ribbon plan, supplements, virtuous reworks, agent-generated harvests, standards-window/runtime harvests, notebook-derived bundles, archive cards, online-source rows, and D-drive source-gap items.	Mine for definitions, evidence, examples, obligations, and alternate phrasings; route every artifact into method, physics-redefinition, application, validator, comparator, or future-expansion lanes.
Transient source	Agent reasoning notes, exploratory hypotheses, analogies, proposed adapters, future derivation lanes.	Use as research channels and prestate declarations; require receipt, theorem, calibration, or falsifier before promotion.

3. Core Doctrine

- 1. Analog access.** The analog toolkit is designed as a universal STEM access kit first. It is also a proof aid and overclaim guard, but its long-term purpose is access to STEM reasoning without requiring electricity, internet, computers, or paper.
- 2. Automation guardrails.** Guardrails regulate AI agents, validators, build scripts, generated prose, and claim-promotion workflows. They are not restrictions on human creativity, analogy, or hypothesis generation.
- 3. Maximal claim posture.** The manuscript may state the largest intended claim, provided the proven pieces and the obligation channels are explicit.
- 4. Open obligations as channels.** An open obligation declares the current prestate and the next evidence path. It is a research channel, not a suppression label.
- 5. Literal physics direction.** The later translation and synthesis papers are expected to make the machinery literal to physics wherever receipt-bound adapters, standard formalism, calibration, and falsifiers can support that move.

4. Methods

4.1 LCR Carrier And Finite Enumeration

The base method treats local state as an ordered LCR carrier with a center and two boundary slots. Early papers establish finite enumeration, shell grading, reversal, and correction-surface accounting. Finite exhaustive enumeration proves only the finite property stated, but it provides a stable substrate for later lift, translation, and adapter work.

4.2 Claim Lanes And Receipts

Each strong claim is routed through a lane: receipt-bound internal result, standard theorem/citation-bound result, CAM/crystal reapplication result, normal-form result, external calibration result, grand synthesis claim with dependencies, or falsifier/open obligation. This is the publication mechanism that lets grand claims be stated without silent evidence promotion.

4.3 Analog Toolkit

The workbook layer converts formal objects into replayable exercises. The target is not only pedagogy; it is robust access. A valid analog kit should preserve the addressed state under lower-tech constraints: no electricity, no internet, no computer, and eventually no paper.

4.4 CAM, Crystals, And Forges

CAM and crystal systems provide addressable memory, projected evidence surfaces, and reusable query/composition patterns. Forge systems actionize CAM into lookup contracts, executable methods, cached answers, event rows, and receipts. The publication treats these as operational surfaces, not decorative terminology.

4.5 TarPit, Bondedness, And Recursive Folding

TarPit is used as a shared metric substrate: data forms are assembled as LCR items, decomposed, read for mass/bond metrics, compared through crystal forms, and then recursively folded as emitted forms. Damascus folding records whether emitted forms remain readable and whether stabilization signals appear across folds.

4.6 Standard Model Definition And Translation

The SMD/SMB layer defines the external Standard Model surfaces that FLCR-31 through FLCR-40 must cite. Translation papers should stop treating every Standard Model-facing row as unresolved by default; rows close when standard formalism, FLCR proof, CAM/crystal reapplication, normal-form translation, calibration, or falsifier protocol satisfies the lane.

5. Paper Architecture

ID	Paper	Role in the unified manuscript	Assembly status	Source status	Receipts
FLCR-01	Grounding Contract And Axiom Discipline	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	5
FLCR-02	The LCR Carrier	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	5
FLCR-03	Correction Surface And Residual Accounting	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	4
FLCR-04	D4, J3(O), Triality, And Representation Transport	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	4
FLCR-05	Typed Boundary Repair	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	4
FLCR-06	Oloid Path Carrier And Transport Geometry	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	4
FLCR-07	Causal Links And Obligation Ledgers	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	5
FLCR-08	Discrete-Continuous Bridge Without Physical Overclaim	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	5
FLCR-09	Lattice Closure And Terminal Addresses	Grounding, carrier, correction, atlas, repair, bridge, ledger, and lattice closure.	ASSEMBLE	full source	4
FLCR-10	Temporal Windows And Hamiltonian Readouts	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	5

ID	Paper	Role in the unified manuscript	Assembly status	Source status	Receipts
FLCR-11	Master Receipt And Stack Replay	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	5
FLCR-12	Theory Admission Gates	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	5
FLCR-13	Cellular Automata Prediction Surfaces	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	4
FLCR-14	Quark-Face Algebra Before Standard-Model Translation	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	4
FLCR-15	Curvature As Boundary-Repair Demand	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	4
FLCR-16	Mass Residue And Carrier Accounting	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	4
FLCR-17	Continuum Edge Residuals	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	4
FLCR-18	Exceptional Towers, VOA Routes, And Observer Face Selection	Temporal, receipt, admission, prediction, algebraic, curvature, mass, continuum, and exceptional machinery.	ASSEMBLE	full source	4
FLCR-19	Layer-2 Synthesis Ledger	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	5
FLCR-20	Applied Forge Reader And Descriptor Kernel	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	REROUTE	full source	5
FLCR-21	Materials Candidate Generation	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	5
FLCR-22	Protein Descriptor And Fold-Facing Kernel	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	5
FLCR-23	Finite Game Lattices And Local Rule Automata	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	5

ID	Paper	Role in the unified manuscript	Assembly status	Source status	Receipts
FLCR-24	Energetic Traversal Maps	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	DISCOVERY	full source	4
FLCR-25	Shear, Plasma, And Carrier Horizons	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	4
FLCR-26	Observer Delay And Shared-State Protocols	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	5
FLCR-27	Monster Ceiling And Large-Invariant Boundaries	Synthesis, forge actionization, applied domains, observer delay, and Monster/invariant boundary.	ASSEMBLE	full source	5
FLCR-28	CAM, Crystal Projectors, And MannyAI Runtime	CAM/crystal runtime, corpus ribbon, supervisor cursor, and normal-form intake.	ASSEMBLE	full source	5
FLCR-29	Corpus Ribbon And Retrospective LCR Readout	CAM/crystal runtime, corpus ribbon, supervisor cursor, and normal-form intake.	ASSEMBLE	full source	6
FLCR-30	Supervisor Cursor And Universal Normal-Form Intake	CAM/crystal runtime, corpus ribbon, supervisor cursor, and normal-form intake.	ASSEMBLE	full source	5
FLCR-31	Gauge Groups Translated Into LCR	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	4
FLCR-32	QCD And Color Transport In LCR	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	4
FLCR-33	Electroweak, Higgs, And Mass Residue Translation	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	4
FLCR-34	Electron-Hole-Exciton Accounting	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	4
FLCR-35	GR, Curvature, And Continuum Translation	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	4
FLCR-36	Condensed Matter, Materials, And Metamaterials	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	5
FLCR-37	Plasma, Energy, And Traversal Calibration	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	4
FLCR-38	Observer, Computation, And AI Runtime Translation	Standard Model bridge-preview and comparator layer.	ASSEMBLE	full source; translation/synthesis layer	5
FLCR-39	Falsifiers, Comparators, And Standards Of Evidence	Standard Model bridge-preview and comparator layer.	REROUTE	full source; translation/synthesis layer	5
FLCR-40	Grand Reconstruction Map And Trust-Removal Protocol	Reconstruction map, trust-removal protocol, and handoff into FLCR-41..80.	ASSEMBLE	full source; translation/synthesis layer	10

6. Results

The current integrated results are not only textual. They include executable or generated receipt surfaces.

Result surface	Current receipt-backed finding
TarPit lift tower	40 FLCR lifts, 320 enumeration rows, 320 successful TarPit runs, 1430 bonds, 2860 triads.
Universal process atlas	8 process families, 71 components, 346 pair bonds, 12 compared targets.
Damascus folding	48 form analyses; final fold pair bonds 168, triads 168, crystal family kagome.
Fine-bonding stabilization	Fold 0 pair bonds 120; stable pair bonds 168; stable triads 168; Hamming address 137; PFC2 realization $137 = 120 + 13 + 4$.
Prime-chart Monster renormalization	Top Monster triple [47, 59, 71]; product 196883; McKay row 196884; Pariah/asymmetry values [33, 37, 39, 44, 53, 65].
Runtime standards-window conformance	Standards/window evidence attached.
Resolved-state closure	320 enumeration rows; success rate 1.0; 40 papers promoted to resolved-state contributions at their satisfied lanes.
Unconsidered implementation intake	7 implementation families bound into 18 papers, including typed kernel, SplatForge field, analog workbook, Rule-30 theorem registry, E8/Leech witnesses, render parity, and CEM negative-control evidence.
Broad capability intake	7 non-obvious capability families routed into 27 papers, including GLM CAM crystals, obligation closure, Rule-30 address horizons, centroid process audit, GaussHaus kernel/forge topology, and proof-suite assembly protocol.
Broad capability reapplication	7 capability operators and 4 recursive combinations reviewed; newly solved surfaces include content-addressed evidence routing, auditable obligation promotion, finite Rule-30 address horizons, closed CAM action loops, forge expression grammar, and executable paper-triad criteria.
External orientation and claim granularity	40 papers updated with external-reader orientation, band-specific scope language, claim-granularity tables, and appendix labels for repeated pass material.
Reviewer claim ledgers	120 formal claim rows expanded across 40 papers into evidence requirements, boundaries, and next review actions.
Peer reviewer assessment	40 papers assessed under a briefed-reviewer assumption; series-level judgment is major revision, technically promising.
Reconstruction map architecture	FLCR-40 reframed as a reconstruction map/trust-removal protocol; FLCR-31..39 marked as bridge previews; FLCR-41..80 reserved for observer-framework domain rebuilds.
Recent quantum evidence intake	3 external quantum sources routed into 7 papers as citation-bound, calibration-bound, or falsifier/comparator evidence.
FLCR NIST/review suite	40 FLCR papers checked; 240 adapted NIST-style verdicts; status counts {'PASS': 240}.
Canonical NIST baseline	Canonical 32-paper baseline run recorded with status counts {'PASS': 192}.
MannyAI/Docker review lanes	7 runtime, standards, window/node, deployment, and governance lanes routed from D:/MannyAI and Docker Starter; standards {'PASS': 192, 'FAIL': 0, 'OPEN': 0, 'EXTERNAL_REQUIRED': 0, 'NOT_APPLICABLE': 0}; suite 776/782; glue 95/95; node verify {'OPEN': 0, 'FAIL': 7, 'PASS': 25}.
D-drive inventory paper gap	203642 review candidates compared against the FLCR corpus; 203038 artifacts are absent, basename-only, or weak root-only coverage. Status counts {'not_integrated': 40, 'basename_seen_only': 30, 'root_covered_only': 180172, 'exact_path_integrated': 526, 'basename_seen_under_covered_root': 22796, 'relative_path_integrated': 78}.
Pairwise external evidence packages	39 internal-source/current-journal-paper packages built with correlation claims, adapted NIST pair reviews, validation reviews, crystal faces, paper routes, and obligations.
Paper assembly audit pipeline	446 scored evidence records; assemble 37, reroute/repair 85, discovery 324.
Archive intake	344 archives indexed with 39645 contained entries; archive records now feed source discovery, slot assignment, and CAM crystal memory.
CAM database intake	102 databases inspected read-only; port recommendations {'PORT_FIRST': 41, 'REVIEW_FOR_PORT': 39, 'LOW_SIGNAL_KEEP_INDEXED': 16, 'NO_PORT_YET': 6}.
Publication-agent products	80 formalized slot-paper products generated; 79 assembly-ready, 80 evidence-rich, 0 skeleton.

6.1 Early Formal Base

FLCR-01 through FLCR-03 now explicitly state that guardrails are automation controls, that obligations are research channels, and that analog workbooks are universal-access STEM kits. FLCR-02 and FLCR-03 inherit this posture while preserving their finite LCR carrier and correction-surface roles.

6.2 Fine-Bonding And 1-3-7

The fine-bonding stabilization pass localizes the current derivation target to the first Damascus refold. The observed transition from 120 pair bonds to stable 168 pair bonds/triads is recorded alongside the Hamming 1-3-7 address and PFC2 realization $137 = 120 + 13 + 4$. The physical alpha promotion remains dependent on the no-target-anchor diagonal-removal adapter.

6.3 Monster Prime-Chart Renormalization

The prime-chart pass encodes 29, 30, 31 $\rightarrow$ 31, 33, 37 as an LCR prime-mask lift. It separates the Happy-family Monster-admitted supersingular account from the Pariah/asymmetry bridge account. At 47, 59, 71, the Monster-side ledger keeps 196883 and 196884, while off-chain bridge values remain a separate asymmetry body.

7. Discussion

The manuscript's central organizational move is to let ambition and rigor coexist. The publication should not reduce the claim to only what is currently easy to prove locally. It should state the intended physical and mathematical direction, attach the proofs already present, and make the missing adapters or calibrations visible as the next channels of work.

This is especially important for the literal-physics direction. Early papers may only prove finite LCR-native structure, but later papers are permitted and expected to work toward literal physical interpretation. The transition must occur through explicit adapters, accepted formalisms, receipts, calibration rows, and falsifiers.

8. Limitations And Open Obligations

The master manuscript preserves the following limitations as active channels:

- Unit-bearing physical adapters must avoid hidden target anchoring.
- Standard Model translation claims must cite SMD definitions before mapping to LCR.
- CAM/crystal reapplication must preserve source identities and receipt lanes.
- Analog kits must be made increasingly usable outside computer-dependent formats.
- Physics literalization must distinguish structural identity, calibrated comparison, and measured-world prediction.

9. Reproducibility And Data Availability

Publication source package: canonical FLCR series tree.

Unified flat export package: final shareable paper folder.

Share archive: final flat export zip.

The flat export includes concatenated formal papers, companion papers, workbooks, receipt passes, generated PDFs, manifests, and this master manuscript.

10. Peer-Review Readiness Checklist

Requirement	Current state	Next action
Unified thesis	Present in this master manuscript	Continue tightening abstract/introduction as claims stabilize.
Source routing	Present through paper manifests and master accounting	Add citation-grade bibliography entries.
Methods	Present but distributed	Convert high-value receipts into figures/tables.
Results	Receipt-backed surfaces present	Add final residual tables and uncertainty rows.
Limitations	Present as obligations/channels	Assign owners/adapters for remaining unit-bearing claims.
PDF submission form	Partial	Finalize LaTeX/arXiv build path and figure package.

11. Conclusion

The FLCR series is now being treated as one integrated scientific publication program. The paper parts remain modular, but the master manuscript supplies the coherent peer-review surface. The next editorial work is to continue replacing local skeleton sections with receipt-backed prose, figures, tables, definitions, and residual ledgers while keeping the full claim posture visible.

Appendix A. Paper Abstract Ledger

FLCR-01 - Grounding Contract And Axiom Discipline

This paper establishes the publication contract for the FLCR series. The core contribution is a conservative-extension discipline: imported mathematics remains external mathematics, while LCR contributes only declared bridges, receipt-bearing finite constructions, and typed claim lanes. The result is a burden system that allows maximal claims to be stated without silently changing their evidentiary type.

FLCR-02 - The LCR Carrier

This paper defines the LCR carrier as the canonical radius-1 local readout with a distinguished center and two ordered boundary slots. The carrier is finite, inspectable, and minimal: any smaller carrier collapses either the center or one boundary address. The paper proves the binary eight-state surface, reversal action, shell grading, and shell-2 doublet-plus-pivot structure used by later papers.

FLCR-03 - Correction Surface And Residual Accounting

This paper formalizes the correction surface as an exact Boolean residual between a linear radius-1 carrier and the Rule 30 target. Over $\text{GF}(2)$, Rule 30 decomposes into Rule 90 plus the residual C and not R. The correction surface is therefore a finite syndrome-like obstruction, not an unexplained repair force and not a physical field without later calibration.

FLCR-04 - D4, J3(O), Triality, And Representation Transport

This paper registers the eight-state LCR carrier across multiple lossless local presentations: LCR triples, an axis/sheet codec, and a diagonal J3(O)-style carrier. The result is a finite coordinate atlas, not a claim that full exceptional dynamics have been solved. Later papers may import representation transport only by preserving the registered readouts or declaring the dropped readout.

FLCR-05 - Typed Boundary Repair

This paper turns failed joins into typed, replayable repair rows. Boundary repair is not a proof of the desired downstream claim; it is a typed exception and proof-obligation discipline that preserves state, coordinate, cause, route, source, target, and falsifier. The main result is an idempotent normalization of failure into legal next-step data.

FLCR-06 - Oloid Path Carrier And Transport Geometry

This paper defines the path carrier as a deterministic finite-state transducer. A repair row can travel as side-channel metadata while the carrier transition remains unchanged. The path theorem closes graph-continuous finite transport and payload noninterference; it does not by itself assert real oloid kinematics, gauge holonomy, or physical motion.

FLCR-07 - Causal Links And Obligation Ledgers

This paper turns the FLCR corpus into an auditable dependency graph. It distinguishes proof-support edges, obligation edges, negative verdicts, receipt edges, and workflow routes. This prevents open obligations from being consumed as proofs while still

allowing them to act as springboards for later reapplication.

FLCR-08 - Discrete-Continuous Bridge Without Physical Overclaim

This paper defines a bridge from finite receipt data to continuous presentation and back to discrete representative form. Piecewise linear interpolation preserves samples; it does not create dynamics. Retraction-style admission into MDHG, SpeedLight, or CAM surfaces must preserve provenance, error budgets, and idempotence.

FLCR-09 - Lattice Closure And Terminal Addresses

This paper installs the lattice and code ladder as an addressable terminal surface. It distinguishes construction claims, lookup claims, invariant claims, glue claims, and physical-mapping claims. The strongest local result is that lattice-forge and CAM surfaces can provide no-cost lookup routes for E8, Niemeier, Leech-facing, and related terminal addresses once receipts are attached.

FLCR-10 - Temporal Windows And Hamiltonian Readouts

This paper turns static carried centers into ordered window objects. A Hamiltonian readout is a finite sliding-window transform with source indices, source centers, forward receipts, backward receipts, and emitted composite centers. The paper closes finite window emergence and kappa-ledger ordering while keeping physical time and unbounded McKay/O2-prime exactness in later claim lanes.

FLCR-11 - Master Receipt And Stack Replay

This paper formalizes the first ten papers as one replayable causal unit. The operative object is T10 master receipt. The core result is that Papers 00-09 can be composed into an inspectable receipt stack when every imported claim preserves source, edge, receipt, and obligation fields. The paper also defines how this result routes forward: FLCR-11 becomes the intake gate for aggregate proof replay and later layer-2 synthesis. Its main residue is explicit: exceptional-to-Niemeier terminal routes and cold extraction claims remain scoped until their own receipts are attached.

FLCR-12 - Theory Admission Gates

This paper formalizes admission control for theories and candidate claims. The operative object is admission gate. The core result is that a candidate theory is admissible only when its object, lane, evidence class, residue, and falsifier are declared before promotion. The paper also defines how this result routes forward: later Standard Model translations must pass through this gate before appearing as FLCR claims. Its main residue is explicit: sporadic-boundary and encoder-invariance examples remain import routes, not blanket admission.

FLCR-13 - Cellular Automata Prediction Surfaces

This paper formalizes bounded prediction surfaces for elementary cellular automata. The operative object is CA prediction surface. The core result is that finite and bounded CA readouts can be receipt-closed without claiming unbounded Rule 30 future-bit extraction. The paper also defines how this result routes forward: finite surfaces feed admission, window, and falsifier papers as testable local machines. Its main residue is explicit: unbounded nonperiodicity, cold nth-bit extraction, and full prediction remain separate mathematical obligations.

FLCR-14 - Quark-Face Algebra Before Standard-Model Translation

This paper formalizes face transport in the finite LCR/J3(O) chart without Standard Model identity language. The operative object is quark-face algebra. The core result is that six excited chart faces admit exact internal transport under the declared finite atlas and closure receipts. The paper also defines how this result routes forward: FLCR-31 and FLCR-32 may translate the structure into gauge/QCD language only after citing this LCR-native base. Its main residue is explicit: measured quark identity, CKM data, and physical color calibration are deferred to translation papers.

FLCR-15 - Curvature As Boundary-Repair Demand

This paper formalizes curvature-like behavior as accumulated repair demand. The operative object is repair-curvature map. The core result is that boundary repair demand can be represented as a structured curvature-like ledger over discrete transitions. The paper also defines how this result routes forward: FLCR-35 may translate this into GR-facing language after preserving the repair-row evidence chain. Its main residue is explicit: Einstein-equation identity, measured curvature, and physical spacetime calibration remain downstream.

FLCR-16 - Mass Residue And Carrier Accounting

This paper formalizes mass-like residue as carrier accounting before QFT translation. The operative object is mass-residue carrier. The core result is that finite sector residues can be carried as internal accounting quantities with explicit source and sector labels. The paper also defines how this result routes forward: FLCR-33 may translate the accounting into Higgs/QFT language only with citation and calibration boundaries. Its main residue is explicit: measured Higgs mass, QFT field identity, and physical mass calibration are not closed locally.

FLCR-17 - Continuum Edge Residuals

This paper formalizes residual behavior at discrete-continuous edges. The operative object is continuum-edge residual. The core result is that edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit. The paper also defines how this result routes forward: later GR, condensed-matter, and calibration papers may consume the edge rows with explicit error budgets. Its main residue is explicit: global continuum collapse and unbounded McKay/O2-prime closure remain separated residues.

FLCR-18 - Exceptional Towers, VOA Routes, And Observer Face Selection

This paper formalizes exceptional tower, VOA, Moonshine, and observer-face routes as bounded LCR machinery. The operative object is exceptional observer route. The core result is that bounded exceptional, VOA, and observer-face readouts can be staged as receipt-bearing routes over the prior LCR carrier. The paper also defines how this result routes forward: FLCR-38 and FLCR-40 may use these routes only after preserving bounded-vs-unbounded status. Its main residue is explicit: full Moonshine identification, measured observer physics, and unbounded exceptional closure remain separate obligations.

FLCR-19 - Layer-2 Synthesis Ledger

This paper formalizes the first higher-layer synthesis ledger over prior papers. The operative object is layer-2 synthesis ledger. The core result is that prior receipt-bearing claims can be assembled into a monotone ledger when edge type and claim lane are preserved. The paper also defines how this result routes forward: this ledger becomes the bridge from base formalization to applied forge papers. Its main residue is explicit: unknown and forbidden reachability must remain explicit rather than hidden in summary prose.

FLCR-20 - Applied Forge Reader And Descriptor Kernel

This paper formalizes the applied forge reader as a descriptor kernel over LCR receipts. The operative object is forge descriptor kernel. The core result is that applied forge descriptors can read, combine, and route LCR-addressed objects without changing the underlying CAM claim state. The paper also defines how this result routes forward: materials, proteins, games, and energy papers consume this as the applied reader interface. Its main residue is explicit: domain performance, external benchmarks, and real-world utility remain external validation tasks.

FLCR-21 - Materials Candidate Generation

This paper formalizes materials candidate generation through forge descriptors and CAM addresses. The operative object is materials forge. The core result is that the internal representation and candidate-generation path can be receipt-bound before fabrication or finite-element validation. The paper also defines how this result routes forward: FLCR-36 may translate this into condensed-matter/materials language with external datasets. Its main residue is explicit: fabrication, finite-element performance, measured band data, and material claims require external calibration.

FLCR-22 - Protein Descriptor And Fold-Facing Kernel

This paper formalizes protein fold descriptors as an applied LCR-facing kernel. The operative object is fold descriptor kernel. The core result is that contact, homology, and winding descriptors can be staged as internal addressable features before biological validation. The paper also defines how this result routes forward: applied biology workbooks may consume descriptors only with PDB/parser and benchmark receipts. Its main residue is explicit: native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind.

FLCR-23 - Finite Game Lattices And Local Rule Automata

This paper formalizes finite game lattices as local rule automata. The operative object is game lattice automaton. The core result is that finite move rules and n-dimensional count surfaces can be closed internally where the board, dimension, and local rule are declared. The paper also defines how this result routes forward: game and combinatorics papers may consume finite counts as receipt-bound local automata. Its main residue is explicit: complete game theory, playability, OEIS review, and real-piece geometry remain external review lanes.

FLCR-24 - Energetic Traversal Maps

This paper formalizes energy-like traversal accounting over LCR paths. The operative object is energetic traversal map. The core result is that internal traversal ledgers can order and compare path costs under declared scalar rules. The paper also defines how this result routes forward: FLCR-37 may translate this into energy/plasma language only with units and measurement lanes. Its main residue is explicit: joule conversion, thermodynamic identity, and physical energy claims require calibration.

FLCR-25 - Shear, Plasma, And Carrier Horizons

This paper formalizes shear and horizon events as carrier-level thresholds. The operative object is carrier horizon. The core result is that pinch, shear, and horizon conditions can be represented as internal carrier threshold events. The paper also defines how this result routes forward: FLCR-37 may use these thresholds for plasma/energy translation after measurement binding. Its main residue is explicit: measured Z-pinch observables, plasma identity, and laboratory calibration remain external.

FLCR-26 - Observer Delay And Shared-State Protocols

This paper formalizes observer delay as a finite shared-state synchronization protocol. The operative object is observer-delay protocol. The core result is that finite observer buffers can synchronize shared center state without implying consciousness or physical collapse. The paper also defines how this result routes forward: FLCR-38 may translate this into AI/computation/observer language with explicit measurement boundaries. Its main residue is explicit: human latency, consciousness, and physical collapse claims require external empirical evidence.

FLCR-27 - Monster Ceiling And Large-Invariant Boundaries

This paper formalizes large invariant ceilings and Monster-facing boundary routes. The operative object is Monster ceiling. The core result is that bounded Monster/McKay invariant facts can function as upper-bound and routing data without closing universal physical energy. The paper also defines how this result routes forward: FLCR-40 may use the ceiling only with explicit dependency and falsifier chains. Its main residue is explicit: measured universal energy bounds, encoder invariance, and full Moonshine identity remain separate obligations.

FLCR-28 - CAM, Crystal Projectors, And MannyAI Runtime

This paper formalizes CAM/crystal projector mechanics and runtime intake. The operative object is crystal projector. The core result is that CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract. The paper also defines how this result routes forward: this paper becomes the runtime intake lane for agent-generated, standards-window, runtime, notebook-derived, and future crystal work. Its main residue is explicit: private or missing CAM vault data must remain marked as pending rather than assumed.

FLCR-29 - Corpus Ribbon And Retrospective LCR Readout

This paper formalizes the whole corpus as a retrospective LCR ribbon. The operative object is corpus ribbon. The core result is that the corpus can be read as an enacted LCR ribbon when each paper's center, routes, receipts, and obligations are preserved. The paper also defines how this result routes forward: FLCR-40 may use this as synthesis evidence only after all dependencies are attached. Its main residue is explicit: retrospective coherence is not a substitute for missing local receipts.

FLCR-30 - Supervisor Cursor And Universal Normal-Form Intake

This paper formalizes supervisor cursor scheduling and universal normal-form intake. The operative object is supervisor cursor. The core result is that a scheduler can traverse paper windows and reserve universal normal-form slots without finalizing claims prematurely. The paper also defines how this result routes forward: FLCR-28 through FLCR-40 use this as the normal-form intake gate. Its main residue is explicit: the incoming universal normal form remains required evidence before final unification closure.

FLCR-31 - Gauge Groups Translated Into LCR

This paper formalizes gauge-group translation from mature LCR-native results. The operative object is gauge translation. The core result is that gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited. The paper also defines how this result routes forward: this paper opens the Standard Model translation band and depends on FLCR-01 through FLCR-14. Its main residue is explicit: measured gauge coupling values and physical identity require citation/calibration lanes.

FLCR-32 - QCD And Color Transport In LCR

This paper formalizes QCD/color language as a translation of finite face transport. The operative object is QCD color translation. The core result is that finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD. The paper also defines how this result routes forward: FLCR-32 cites FLCR-14 before making any color-transport claim. Its main residue is explicit: physical quark confinement, CKM, and measured QCD parameters remain external calibration.

FLCR-33 - Electroweak, Higgs, And Mass Residue Translation

This paper formalizes electroweak/Higgs language as translation of mass-residue accounting. The operative object is Higgs residue translation. The core result is that mass-residue carrier accounting can be translated into electroweak/Higgs-facing terms only when sector, units, and calibration are declared. The paper also defines how this result routes forward: FLCR-33 cites FLCR-16 and standard references before physical mass claims. Its main residue is explicit: Higgs VEV, measured masses, and QFT field identity require external data binding.

FLCR-34 - Electron-Hole-Exciton Accounting

This paper formalizes electron-hole-exciton formalism as standard-model-adjacent accounting for residues. The operative object is exciton accounting. The core result is that standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared. The paper also defines how this result routes forward: FLCR-34 supplies the missing condensed-matter bridge for papers that need carrier-pair accounting. Its main residue is explicit: material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration.

FLCR-35 - GR, Curvature, And Continuum Translation

This paper formalizes general-relativity and continuum language as translation of repair-curvature and edge residuals. The operative object is GR-continuum translation. The core result is that discrete repair-curvature and continuum-edge rows can be translated into GR-facing claims only when metric, limit, and calibration assumptions are explicit. The paper also defines how this result routes forward: FLCR-35 cites FLCR-15 and FLCR-17 as its LCR-native base. Its main residue is explicit: Einstein-field-equation identity and measured spacetime curvature remain external calibration.

FLCR-36 - Condensed Matter, Materials, And Metamaterials

This paper formalizes condensed-matter/materials translation of forge candidates and exciton accounting. The operative object is materials translation. The core result is that forge-generated material candidates can be translated into condensed-matter hypotheses when descriptors, band context, and validation lane are attached. The paper also defines how this result routes forward: FLCR-36 ties FLCR-21, FLCR-34, and external materials data into one validation map. Its main residue is explicit: fabrication, finite-element performance, spectroscopy, and device claims remain empirical.

FLCR-37 - Plasma, Energy, And Traversal Calibration

This paper formalizes plasma and energy translation from traversal ledgers and carrier horizons. The operative object is energy-plasma translation. The core result is that internal traversal cost and carrier thresholds can be translated into plasma/energy hypotheses only with unit conversion and measured observables. The paper also defines how this result routes forward: FLCR-37 composes FLCR-24, FLCR-25, and FLCR-27 with calibration boundaries. Its main residue is explicit: laboratory plasma behavior, joule conversion, and universal energy bounds require external data.

FLCR-38 - Observer, Computation, And AI Runtime Translation

This paper formalizes observer, computation, and AI runtime translation of LCR state selection. The operative object is observer-runtime translation. The core result is that observer-face selection, delay buffers, and corpus ribbons can be translated into AI/runtime protocols while preserving local receipt state. The paper also defines how this result routes forward: FLCR-38 connects the LCR-native observer papers to runtime, CAM, and runtime orchestration. Its main residue is explicit: consciousness, human perception, and physical collapse claims remain empirical or philosophical unless separately evidenced.

FLCR-39 - Falsifiers, Comparators, And Standards Of Evidence

This paper formalizes series-wide falsifiers, comparators, and evidence standards. The operative object is evidence comparator. The core result is that every FLCR claim can be evaluated by lane, source, receipt, comparator, and falsifier before final admission. The paper also defines how this result routes forward: FLCR-39 supplies the standards suite for the final unification paper. Its main residue is explicit: claims that lack comparator or falsifier fields remain obligations regardless of rhetorical strength.

FLCR-40 - Grand Reconstruction Map And Trust-Removal Protocol

This paper is the reconstruction map for the first forty FLCR papers. It does not introduce a new receipt-bearing proof of the full synthesis. Instead, it gives the dependency map, trust-removal protocol, analog reconstruction route, and tool/code correspondence needed for a reader to reconstruct or reject the FLCR claim without accepting the author's terminology in advance. Prior receipts remain in the papers, supplements, CAM/crystal records, and code/tool runners that produced them. FLCR-40 routes those materials and marks every missing receipt, comparator, calibration, or dependency as a blocker. The paper also defines the handoff into FLCR-41 through FLCR-80, where the observer/readout framework is applied domain-by-domain to Standard Model and adjacent physics structures.

Appendix B. Formal Claim Blocks

FLCR-01 - Grounding Contract And Axiom Discipline

Claim	Lane	Statement
Theorem 1.1	normal_form_result	The FLCR corpus is admissible only as a typed extension stack: every promoted claim must declare its lane, source, evidence, and boundary.
Proposition 1.2	standard_theorem_citation_bound_result	For finite domains such as the eight binary LCR states, exhaustive enumeration is complete proof of the enumerated property.
Protocol 1.3	falsifier_or_open_obligation	A claim without a receipt, citation, calibration datum, CAM route, or falsifier boundary remains a staged obligation.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 1.1	formal construction	The FLCR corpus is admissible only as a typed extension stack: every promoted claim must declare its lane, source, evidence, and boundary	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 1.2	formal construction	For finite domains such as the eight binary LCR states, exhaustive enumeration is complete proof of the enumerated property	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Protocol 1.3	obligation/falsifier	A claim without a receipt, citation, calibration datum, CAM route, or falsifier boundary remains a staged obligation	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-01 defines or uses Grounding Contract And Axiom Discipline as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-02 - The LCR Carrier

Claim	Lane	Statement
Theorem 2.1	receipt_bound_internal_result	The LCR carrier is the minimal address-preserving carrier for one center and two distinguishable boundary slots.
Theorem 2.2	receipt_bound_internal_result	In the binary chart, reversal is an involution and partitions the carrier into fixed states and reversal pairs.
Proposition 2.3	standard_theorem_citation_bound_result	The shell profile of $\{0,1\}^3$ is the binomial profile 1,3,3,1.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 2.1	finite/executable result	The LCR carrier is the minimal address-preserving carrier for one center and two distinguishable boundary slots	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Theorem 2.2	finite/executable result	In the binary chart, reversal is an involution and partitions the carrier into fixed states and reversal pairs	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 2.3	formal construction	The shell profile of $\{0,1\}^3$ is the binomial profile 1,3,3,1	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-02 defines or uses The LCR Carrier as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-03 - Correction Surface And Residual Accounting

Claim	Lane	Statement
Theorem 3.1	receipt_bound_internal_result	The Rule 30 correction surface over the binary LCR chart is exactly C and not R.
Proposition 3.2	receipt_bound_internal_result	The left bit indexes two copies of the same obstruction because the residual condition is independent of L.
Protocol 3.3	falsifier_or_open_obligation	Any later physical, McKay, VOA, or telemetry interpretation must consume this residual through a typed promotion lane.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 3.1	finite/executable result	The Rule 30 correction surface over the binary LCR chart is exactly C and not R	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 3.2	finite/executable result	The left bit indexes two copies of the same obstruction because the residual condition is independent of L	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 3.3	obligation/falsifier	Any later physical, McKay, VOA, or telemetry interpretation must consume this residual through a typed promotion lane	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-03 defines or uses Correction Surface And Residual Accounting as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-04 - D4, J3(O), Triality, And Representation Transport

Claim	Lane	Statement
Theorem 4.1	receipt_bound_internal_result	The eight binary LCR states admit a lossless finite atlas across LCR, axis/sheet, and diagonal carrier coordinates.
Proposition 4.2	normal_form_result	The axis/sheet codec is an antipodal quotient followed by a sheet lift.
Protocol 4.3	falsifier_or_open_obligation	Full D4, F4, or off-diagonal J3(O) assertions are downstream claims unless a separate action receipt is attached.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 4.1	finite/executable result	The eight binary LCR states admit a lossless finite atlas across LCR, axis/sheet, and diagonal carrier coordinates	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 4.2	formal construction	The axis/sheet codec is an antipodal quotient followed by a sheet lift	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 4.3	obligation/falsifier	Full D4, F4, or off-diagonal J3(O) assertions are downstream claims unless a separate action receipt is attached	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-04 defines or uses D4, J3(O), Triality, And Representation Transport as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-05 - Typed Boundary Repair

Claim	Lane	Statement
Theorem 5.1	receipt_bound_internal_result	A valid boundary repair converts a failed join into a typed proof-obligation row while preserving the relevant coordinates.
Proposition 5.2	receipt_bound_internal_result	Repair is idempotent on already-normalized rows.
Protocol 5.3	normal_form_result	Untyped failure cannot be consumed by later papers as repair evidence.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 5.1	finite/executable result	A valid boundary repair converts a failed join into a typed proof-obligation row while preserving the relevant coordinates	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 5.2	finite/executable result	Repair is idempotent on already-normalized rows	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 5.3	formal construction	Untyped failure cannot be consumed by later papers as repair evidence	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-05 defines or uses Typed Boundary Repair as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-06 - Oloid Path Carrier And Transport Geometry

Claim	Lane	Statement
Theorem 6.1	receipt_bound_internal_result	The path carrier is a deterministic finite-state transducer over its declared state space.
Theorem 6.2	receipt_bound_internal_result	Repair-row payloads are noninterfering side-channel data for the carrier transition.
Protocol 6.3	falsifier_or_open_obligation	Physical oloid, holonomy, QCD, or mass interpretations require downstream calibration or theorem import.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 6.1	finite/executable result	The path carrier is a deterministic finite-state transducer over its declared state space	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Theorem 6.2	finite/executable result	Repair-row payloads are noninterfering side-channel data for the carrier transition	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 6.3	obligation/falsifier	Physical oloid, holonomy, QCD, or mass interpretations require downstream calibration or theorem import	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-06 defines or uses Oloid Path Carrier And Transport Geometry as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-07 - Causal Links And Obligation Ledgers

Claim	Lane	Statement
Theorem 7.1	<code>normal_form_result</code>	A valid FLCR causal graph must type its edges before they can be consumed by paper claims.
Proposition 7.2	<code>receipt_bound_internal_result</code>	Negative extraction verdicts reduce future search space and therefore belong in the ledger.
Protocol 7.3	<code>falsifier_or_open_obligation</code>	Obligation edges may route future work but cannot serve as proof-support edges without promotion.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 7.1	formal construction	A valid FLCR causal graph must type its edges before they can be consumed by paper claims	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 7.2	finite/executable result	Negative extraction verdicts reduce future search space and therefore belong in the ledger	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 7.3	obligation/falsifier	Obligation edges may route future work but cannot serve as proof-support edges without promotion	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-07 defines or uses Causal Links And Obligation Ledgers as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-08 - Discrete-Continuous Bridge Without Physical Overclaim

Claim	Lane	Statement
Theorem 8.1	standard_theorem_citation_bound_result	Every finite numeric trace admits a piecewise-linear interpolant preserving declared samples.
Theorem 8.2	normal_form_result	A valid discrete-continuous bridge preserves sample identity and provenance but not between-sample dynamics by default.
Protocol 8.3	external_calibration_result	Physical or numerical dynamics claims require an equation, sampling model, error norm, or measured trace receipt.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 8.1	formal construction	Every finite numeric trace admits a piecewise-linear interpolant preserving declared samples	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Theorem 8.2	formal construction	A valid discrete-continuous bridge preserves sample identity and provenance but not between-sample dynamics by default	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 8.3	formal construction	Physical or numerical dynamics claims require an equation, sampling model, error norm, or measured trace receipt	dataset, measured value, uncertainty, comparator, calibration protocol, or external benchmark	only the calibrated target, scale, units, and comparator stated in the row	attach dataset/citation, uncertainty, pass/fail criterion, and falsifier

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-08 defines or uses Discrete-Continuous Bridge Without Physical Overclaim as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-09 - Lattice Closure And Terminal Addresses

Claim	Lane	Statement
Theorem 9.1	cam_crystal_reapplication_result	The FLCR lattice ladder supports receipt-bound terminal lookup across the declared code/lattice chain.
Proposition 9.2	receipt_bound_internal_result	No-cost Leech lookup is a valid computational capability once the lookup receipt is attached, but it is not a proof of every Leech invariant.
Protocol 9.3	falsifier_or_open_obligation	Nontrivial glue, Gamma72, orbit, and physical-map claims require their own evidence lanes.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 9.1	CAM route	The FLCR lattice ladder supports receipt-bound terminal lookup across the declared code/lattice chain	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 9.2	finite/executable result	No-cost Leech lookup is a valid computational capability once the lookup receipt is attached, but it is not a proof of every Leech invariant	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 9.3	obligation/falsifier	Nontrivial glue, Gamma72, orbit, and physical-map claims require their own evidence lanes	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-09 defines or uses Lattice Closure And Terminal Addresses as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-10 - Temporal Windows And Hamiltonian Readouts

Claim	Lane	Statement
Theorem 10.1	standard_theorem_citation_bound_result	A finite sequence of n center states has exactly n - w + 1 width-w Hamiltonian windows when w <= n.
Theorem 10.2	receipt_bound_internal_result	Each admitted window preserves source index, source center, forward receipt, backward receipt, and emitted composite center.
Protocol 10.3	falsifier_or_open_obligation	McKay threshold exactness and physical-time interpretation are not granted by finite window admissibility alone.

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 10.1	formal construction	A finite sequence of n center states has exactly n - w + 1 width-w Hamiltonian windows when w <= n	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Theorem 10.2	finite/executable result	Each admitted window preserves source index, source center, forward receipt, backward receipt, and emitted composite center	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Protocol 10.3	obligation/falsifier	McKay threshold exactness and physical-time interpretation are not granted by finite window admissibility alone	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-10 defines or uses Temporal Windows And Hamiltonian Readouts as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-11 - Master Receipt And Stack Replay

Claim	Lane	Statement
Theorem 11.1	receipt_bound_internal_result	Papers 00-09 can be composed into an inspectable receipt stack when every imported claim preserves source, edge, receipt, and obligation fields
Proposition 11.2	normal_form_result	T10 master receipt can be imported by later papers only through its declared source, receipt, and lane.
Protocol 11.3	falsifier_or_open_obligation	exceptional-to-Niemeier terminal routes and cold extraction claims remain scoped until their own receipts are attached

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 11.1	finite/executable result	Papers 00-09 can be composed into an inspectable receipt stack when every imported claim preserves source, edge, receipt, and obligation fields	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 11.2	formal construction	T10 master receipt can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Protocol 11.3	obligation/falsifier	exceptional-to-Niemeier terminal routes and cold extraction claims remain scoped until their own receipts are attached	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-11 defines or uses Master Receipt And Stack Replay as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-12 - Theory Admission Gates

Claim	Lane	Statement
Theorem 12.1	receipt_bound_internal_result	a candidate theory is admissible only when its object, lane, evidence class, residue, and falsifier are declared before promotion
Proposition 12.2	normal_form_result	admission gate can be imported by later papers only through its declared source, receipt, and lane.
Protocol 12.3	falsifier_or_open_obligation	sporadic-boundary and encoder-invariance examples remain import routes, not blanket admission

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 12.1	finite/executable result	a candidate theory is admissible only when its object, lane, evidence class, residue, and falsifier are declared before promotion	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Proposition 12.2	formal construction	admission gate can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 12.3	obligation/falsifier	sporadic-boundary and encoder-invariance examples remain import routes, not blanket admission	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-12 defines or uses Theory Admission Gates as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-13 - Cellular Automata Prediction Surfaces

Claim	Lane	Statement
Theorem 13.1	receipt_bound_internal_result	finite and bounded CA readouts can be receipt-closed without claiming unbounded Rule 30 future-bit extraction
Proposition 13.2	normal_form_result	CA prediction surface can be imported by later papers only through its declared source, receipt, and lane.
Protocol 13.3	falsifier_or_open_obligation	unbounded nonperiodicity, cold nth-bit extraction, and full prediction remain separate mathematical obligations

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 13.1	finite/executable result	finite and bounded CA readouts can be receipt-closed without claiming unbounded Rule 30 future-bit extraction	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 13.2	formal construction	CA prediction surface can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 13.3	obligation/falsifier	unbounded nonperiodicity, cold nth-bit extraction, and full prediction remain separate mathematical obligations	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-13 defines or uses Cellular Automata Prediction Surfaces as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-14 - Quark-Face Algebra Before Standard-Model Translation

Claim	Lane	Statement
Theorem 14.1	receipt_bound_internal_result	six excited chart faces admit exact internal transport under the declared finite atlas and closure receipts
Proposition 14.2	normal_form_result	quark-face algebra can be imported by later papers only through its declared source, receipt, and lane.
Protocol 14.3	falsifier_or_open_obligation	measured quark identity, CKM data, and physical color calibration are deferred to translation papers

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 14.1	finite/executable result	six excited chart faces admit exact internal transport under the declared finite atlas and closure receipts	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 14.2	formal construction	quark-face algebra can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 14.3	obligation/falsifier	measured quark identity, CKM data, and physical color calibration are deferred to translation papers	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-14 defines or uses Quark-Face Algebra Before Standard-Model Translation as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-15 - Curvature As Boundary-Repair Demand

Claim	Lane	Statement
Theorem 15.1	receipt_bound_internal_result	boundary repair demand can be represented as a structured curvature-like ledger over discrete transitions
Proposition 15.2	normal_form_result	repair-curvature map can be imported by later papers only through its declared source, receipt, and lane.
Protocol 15.3	falsifier_or_open_obligation	Einstein-equation identity, measured curvature, and physical spacetime calibration remain downstream

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 15.1	finite/executable result	boundary repair demand can be represented as a structured curvature-like ledger over discrete transitions	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 15.2	formal construction	repair-curvature map can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 15.3	obligation/falsifier	Einstein-equation identity, measured curvature, and physical spacetime calibration remain downstream	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-15 defines or uses Curvature As Boundary-Repair Demand as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-16 - Mass Residue And Carrier Accounting

Claim	Lane	Statement
Theorem 16.1	receipt_bound_internal_result	finite sector residues can be carried as internal accounting quantities with explicit source and sector labels
Proposition 16.2	normal_form_result	mass-residue carrier can be imported by later papers only through its declared source, receipt, and lane.
Protocol 16.3	falsifier_or_open_obligation	measured Higgs mass, QFT field identity, and physical mass calibration are not closed locally

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 16.1	finite/executable result	finite sector residues can be carried as internal accounting quantities with explicit source and sector labels	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 16.2	formal construction	mass-residue carrier can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 16.3	obligation/falsifier	measured Higgs mass, QFT field identity, and physical mass calibration are not closed locally	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-16 defines or uses Mass Residue And Carrier Accounting as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-17 - Continuum Edge Residuals

Claim	Lane	Statement
Theorem 17.1	<code>normal_form_result</code>	edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit
Proposition 17.2	<code>normal_form_result</code>	continuum-edge residual can be imported by later papers only through its declared source, receipt, and lane.
Protocol 17.3	<code>falsifier_or_open_obligation</code>	global continuum collapse and unbounded McKay/O2-prime closure remain separated residues

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 17.1	formal construction	edge residuals can be tracked as finite bridge artifacts without asserting a global continuum limit	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 17.2	formal construction	continuum-edge residual can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 17.3	obligation/falsifier	global continuum collapse and unbounded McKay/O2-prime closure remain separated residues	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-17 defines or uses Continuum Edge Residuals as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-18 - Exceptional Towers, VOA Routes, And Observer Face Selection

Claim	Lane	Statement
Theorem 18.1	cam_crystal_reapplication_result	bounded exceptional, VOA, and observer-face readouts can be staged as receipt-bearing routes over the prior LCR carrier
Proposition 18.2	normal_form_result	exceptional observer route can be imported by later papers only through its declared source, receipt, and lane.
Protocol 18.3	falsifier_or_open_obligation	full Moonshine identification, measured observer physics, and unbounded exceptional closure remain separate obligations

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 18.1	CAM route	bounded exceptional, VOA, and observer-face readouts can be staged as receipt-bearing routes over the prior LCR carrier	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 18.2	formal construction	exceptional observer route can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 18.3	obligation/falsifier	full Moonshine identification, measured observer physics, and unbounded exceptional closure remain separate obligations	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-18 defines or uses Exceptional Towers, VOA Routes, And Observer Face Selection as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-19 - Layer-2 Synthesis Ledger

Claim	Lane	Statement
Theorem 19.1	receipt_bound_internal_result	prior receipt-bearing claims can be assembled into a monotone ledger when edge type and claim lane are preserved
Proposition 19.2	normal_form_result	layer-2 synthesis ledger can be imported by later papers only through its declared source, receipt, and lane.
Protocol 19.3	falsifier_or_open_obligation	unknown and forbidden reachability must remain explicit rather than hidden in summary prose

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 19.1	finite/executable result	prior receipt-bearing claims can be assembled into a monotone ledger when edge type and claim lane are preserved	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 19.2	formal construction	layer-2 synthesis ledger can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 19.3	obligation/falsifier	unknown and forbidden reachability must remain explicit rather than hidden in summary prose	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-19 defines or uses Layer-2 Synthesis Ledger as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-20 - Applied Forge Reader And Descriptor Kernel

Claim	Lane	Statement
Theorem 20.1	receipt_bound_internal_result	applied forge descriptors can read, combine, and route LCR-addressed objects without changing the underlying CAM claim state
Proposition 20.2	normal_form_result	forge descriptor kernel can be imported by later papers only through its declared source, receipt, and lane.
Protocol 20.3	falsifier_or_open_obligation	domain performance, external benchmarks, and real-world utility remain external validation tasks

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 20.1	finite/executable result	applied forge descriptors can read, combine, and route LCR-addressed objects without changing the underlying CAM claim state	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 20.2	formal construction	forge descriptor kernel can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 20.3	obligation/falsifier	domain performance, external benchmarks, and real-world utility remain external validation tasks	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-20 defines or uses Applied Forge Reader And Descriptor Kernel as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-21 - Materials Candidate Generation

Claim	Lane	Statement
Theorem 21.1	cam_crystal_reapplication_result	the internal representation and candidate-generation path can be receipt-bound before fabrication or finite-element validation
Proposition 21.2	normal_form_result	materials forge can be imported by later papers only through its declared source, receipt, and lane.
Protocol 21.3	falsifier_or_open_obligation	fabrication, finite-element performance, measured band data, and material claims require external calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 21.1	CAM route	the internal representation and candidate-generation path can be receipt-bound before fabrication or finite-element validation	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 21.2	formal construction	materials forge can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 21.3	obligation/falsifier	fabrication, finite-element performance, measured band data, and material claims require external calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-21 defines or uses Materials Candidate Generation as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-22 - Protein Descriptor And Fold-Facing Kernel

Claim	Lane	Statement
Theorem 22.1	normal_form_result	contact, homology, and winding descriptors can be staged as internal addressable features before biological validation
Proposition 22.2	normal_form_result	fold descriptor kernel can be imported by later papers only through its declared source, receipt, and lane.
Protocol 22.3	falsifier_or_open_obligation	native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 22.1	formal construction	contact, homology, and winding descriptors can be staged as internal addressable features before biological validation	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 22.2	formal construction	fold descriptor kernel can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 22.3	obligation/falsifier	native-structure prediction, fold-rate validation, and biological performance remain below full closure until datasets bind	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-22 defines or uses Protein Descriptor And Fold-Facing Kernel as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-23 - Finite Game Lattices And Local Rule Automata

Claim	Lane	Statement
Theorem 23.1	receipt_bound_internal_result	finite move rules and n-dimensional count surfaces can be closed internally where the board, dimension, and local rule are declared
Proposition 23.2	normal_form_result	game lattice automaton can be imported by later papers only through its declared source, receipt, and lane.
Protocol 23.3	falsifier_or_open_obligation	complete game theory, playability, OEIS review, and real-piece geometry remain external review lanes

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 23.1	finite/executable result	finite move rules and n-dimensional count surfaces can be closed internally where the board, dimension, and local rule are declared	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 23.2	formal construction	game lattice automaton can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 23.3	obligation/falsifier	complete game theory, playability, OEIS review, and real-piece geometry remain external review lanes	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR - 23 defines or uses Finite Game Lattices And Local Rule Automata as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-24 - Energetic Traversal Maps

Claim	Lane	Statement
Theorem 24.1	receipt_bound_internal_result	internal traversal ledgers can order and compare path costs under declared scalar rules
Proposition 24.2	normal_form_result	energetic traversal map can be imported by later papers only through its declared source, receipt, and lane.
Protocol 24.3	falsifier_or_open_obligation	joule conversion, thermodynamic identity, and physical energy claims require calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 24.1	finite/executable result	internal traversal ledgers can order and compare path costs under declared scalar rules	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 24.2	formal construction	energetic traversal map can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 24.3	obligation/falsifier	joule conversion, thermodynamic identity, and physical energy claims require calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-24 defines or uses Energetic Traversal Maps as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-25 - Shear, Plasma, And Carrier Horizons

Claim	Lane	Statement
Theorem 25.1	cam_crystal_reapplication_result	pinch, shear, and horizon conditions can be represented as internal carrier threshold events
Proposition 25.2	normal_form_result	carrier horizon can be imported by later papers only through its declared source, receipt, and lane.
Protocol 25.3	falsifier_or_open_obligation	measured Z-pinch observables, plasma identity, and laboratory calibration remain external

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 25.1	CAM route	pinch, shear, and horizon conditions can be represented as internal carrier threshold events	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 25.2	formal construction	carrier horizon can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 25.3	obligation/falsifier	measured Z-pinch observables, plasma identity, and laboratory calibration remain external	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-25 defines or uses Shear, Plasma, And Carrier Horizons as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-26 - Observer Delay And Shared-State Protocols

Claim	Lane	Statement
Theorem 26.1	receipt_bound_internal_result	finite observer buffers can synchronize shared center state without implying consciousness or physical collapse
Proposition 26.2	normal_form_result	observer-delay protocol can be imported by later papers only through its declared source, receipt, and lane.
Protocol 26.3	falsifier_or_open_obligation	human latency, consciousness, and physical collapse claims require external empirical evidence

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 26.1	finite/executable result	finite observer buffers can synchronize shared center state without implying consciousness or physical collapse	receipt, validator, executable pass, generated artifact, or finite enumeration	the stated finite/executable domain only	verify the receipt exists and its scope matches the claim
Proposition 26.2	formal construction	observer-delay protocol can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 26.3	obligation/falsifier	human latency, consciousness, and physical collapse claims require external empirical evidence	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-26 defines or uses Observer Delay And Shared-State Protocols as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-27 - Monster Ceiling And Large-Invariant Boundaries

Claim	Lane	Statement
Theorem 27.1	cam_crystal_reapplication_result	bounded Monster/McKay invariant facts can function as upper-bound and routing data without closing universal physical energy
Proposition 27.2	normal_form_result	Monster ceiling can be imported by later papers only through its declared source, receipt, and lane.
Protocol 27.3	falsifier_or_open_obligation	measured universal energy bounds, encoder invariance, and full Moonshine identity remain separate obligations

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 27.1	CAM route	bounded Monster/McKay invariant facts can function as upper-bound and routing data without closing universal physical energy	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 27.2	formal construction	Monster ceiling can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 27.3	obligation/falsifier	measured universal energy bounds, encoder invariance, and full Moonshine identity remain separate obligations	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-27 defines or uses Monster Ceiling And Large-Invariant Boundaries as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-28 - CAM, Crystal Projectors, And MannyAI Runtime

Claim	Lane	Statement
Theorem 28.1	cam_crystal_reapplication_result	CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract
Proposition 28.2	normal_form_result	crystal projector can be imported by later papers only through its declared source, receipt, and lane.
Protocol 28.3	falsifier_or_open_obligation	private or missing CAM vault data must remain marked as pending rather than assumed

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 28.1	CAM route	CAM-addressed objects can be projected through multiple representational faces while remaining addressable by the same memory contract	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 28.2	formal construction	crystal projector can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 28.3	obligation/falsifier	private or missing CAM vault data must remain marked as pending rather than assumed	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-28 defines or uses CAM, Crystal Projectors, And Runtime as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-29 - Corpus Ribbon And Retrospective LCR Readout

Claim	Lane	Statement
Theorem 29.1	grand_synthesis_claim	the corpus can be read as an enacted LCR ribbon when each paper's center, routes, receipts, and obligations are preserved
Proposition 29.2	normal_form_result	corpus ribbon can be imported by later papers only through its declared source, receipt, and lane.
Protocol 29.3	falsifier_or_open_obligation	retrospective coherence is not a substitute for missing local receipts

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 29.1	dependency-explicit synthesis	the corpus can be read as an enacted LCR ribbon when each paper's center, routes, receipts, and obligations are preserved	dependency graph over prior papers, receipts, citations, calibration rows, and falsifiers	inherits every dependency boundary; no unconditional universal claim without all dependencies	expand dependency table and list unresolved blockers
Proposition 29.2	formal construction	corpus ribbon can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 29.3	obligation/falsifier	retrospective coherence is not a substitute for missing local receipts	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-29 defines or uses Corpus Ribbon And Retrospective LCR Readout as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-30 - Supervisor Cursor And Universal Normal-Form Intake

Claim	Lane	Statement
Theorem 30.1	normal_form_result	a scheduler can traverse paper windows and reserve universal normal-form slots without finalizing claims prematurely
Proposition 30.2	normal_form_result	supervisor cursor can be imported by later papers only through its declared source, receipt, and lane.
Protocol 30.3	falsifier_or_open_obligation	the incoming universal normal form remains required evidence before final unification closure

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 30.1	formal construction	a scheduler can traverse paper windows and reserve universal normal-form slots without finalizing claims prematurely	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 30.2	formal construction	supervisor cursor can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 30.3	obligation/falsifier	the incoming universal normal form remains required evidence before final unification closure	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-30 defines or uses Supervisor Cursor And Universal Normal-Form Intake as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	Imported mathematics remains an external theorem/citation target; this paper may use it only through declared mappings and conservative-extension discipline.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Physical or Standard Model identity is not claimed here unless the paper contains an explicit calibration row; the default status is deferred mapping.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-31 - Gauge Groups Translated Into LCR

Claim	Lane	Statement
Theorem 31.1	standard_theorem_citation_bound_result	gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited
Proposition 31.2	normal_form_result	gauge translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 31.3	falsifier_or_open_obligation	measured gauge coupling values and physical identity require citation/calibration lanes

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 31.1	external target or comparator	gauge-like structures may be stated as translations only after the LCR-native carrier, atlas, face, and receipt dependencies are cited	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Proposition 31.2	translation grammar	gauge translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 31.3	obligation/falsifier	measured gauge coupling values and physical identity require citation/calibration lanes	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR- 31 defines or uses Gauge Groups Translated Into LCR as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-32 - QCD And Color Transport In LCR

Claim	Lane	Statement
Theorem 32.1	standard_theorem_citation_bound_result	finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD
Proposition 32.2	normal_form_result	QCD color translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 32.3	falsifier_or_open_obligation	physical quark confinement, CKM, and measured QCD parameters remain external calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 32.1	external target or comparator	finite six-face transport can support a QCD-facing translation only with explicit scope separating structural color from measured QCD	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Proposition 32.2	translation grammar	QCD color translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 32.3	obligation/falsifier	physical quark confinement, CKM, and measured QCD parameters remain external calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-32 defines or uses QCD And Color Transport In LCR as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-33 - Electroweak, Higgs, And Mass Residue Translation

Claim	Lane	Statement
Theorem 33.1	external_calibration_result	mass-residue carrier accounting can be translated into electroweak/Higgs-facing terms only when sector, units, and calibration are declared
Proposition 33.2	normal_form_result	Higgs residue translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 33.3	falsifier_or_open_obligation	Higgs VEV, measured masses, and QFT field identity require external data binding

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 33.1	external target or comparator	mass-residue carrier accounting can be translated into electroweak/Higgs-facing terms only when sector, units, and calibration are declared	dataset, measured value, uncertainty, comparator, calibration protocol, or external benchmark	only the calibrated target, scale, units, and comparator stated in the row	attach dataset/citation, uncertainty, pass/fail criterion, and falsifier
Proposition 33.2	translation grammar	Higgs residue translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 33.3	obligation/falsifier	Higgs VEV, measured masses, and QFT field identity require external data binding	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR- 33 defines or uses Electroweak, Higgs, And Mass Residue Translation as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-34 - Electron-Hole-Exciton Accounting

Claim	Lane	Statement
Theorem 34.1	standard_theorem_citation_bound_result	standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared
Proposition 34.2	normal_form_result	exciton accounting can be imported by later papers only through its declared source, receipt, and lane.
Protocol 34.3	falsifier_or_open_obligation	material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 34.1	external target or comparator	standard electron, hole, recombination, screening, and exciton models can explain part of the open residue surface when material context is declared	named external theorem, definition layer, citation target, or imported standard formalism	imports the external object as-is; does not silently reprove or redefine it	attach final citation metadata and mapping rule
Proposition 34.2	translation grammar	exciton accounting can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 34.3	obligation/falsifier	material-specific bands, lifetimes, mobilities, and spectra require data-bound calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-34 defines or uses Electron-Hole-Exciton Accounting as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-35 - GR, Curvature, And Continuum Translation

Claim	Lane	Statement
Theorem 35.1	external_calibration_result	discrete repair-curvature and continuum-edge rows can be translated into GR-facing claims only when metric, limit, and calibration assumptions are explicit
Proposition 35.2	normal_form_result	GR-continuum translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 35.3	falsifier_or_open_obligation	Einstein-field-equation identity and measured spacetime curvature remain external calibration

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 35.1	external target or comparator	discrete repair-curvature and continuum-edge rows can be translated into GR-facing claims only when metric, limit, and calibration assumptions are explicit	dataset, measured value, uncertainty, comparator, calibration protocol, or external benchmark	only the calibrated target, scale, units, and comparator stated in the row	attach dataset/citation, uncertainty, pass/fail criterion, and falsifier
Proposition 35.2	translation grammar	GR-continuum translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 35.3	obligation/falsifier	Einstein-field-equation identity and measured spacetime curvature remain external calibration	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-35 defines or uses GR, Curvature, And Continuum Translation as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-36 - Condensed Matter, Materials, And Metamaterials

Claim	Lane	Statement
Theorem 36.1	external_calibration_result	forge-generated material candidates can be translated into condensed-matter hypotheses when descriptors, band context, and validation lane are attached
Proposition 36.2	normal_form_result	materials translation can be imported by later papers only through its declared source, receipt, and lane.

Claim	Lane	Statement
Protocol 36.3	falsifier_or_open_obligation	fabrication, finite-element performance, spectroscopy, and device claims remain empirical

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 36.1	external target or comparator	forge-generated material candidates can be translated into condensed-matter hypotheses when descriptors, band context, and validation lane are attached	dataset, measured value, uncertainty, comparator, calibration protocol, or external benchmark	only the calibrated target, scale, units, and comparator stated in the row	attach dataset/citation, uncertainty, pass/fail criterion, and falsifier
Proposition 36.2	translation grammar	materials translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 36.3	obligation/falsifier	fabrication, finite-element performance, spectroscopy, and device claims remain empirical	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-36 defines or uses Condensed Matter, Materials, And Metamaterials as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-37 - Plasma, Energy, And Traversal Calibration

Claim	Lane	Statement
Theorem 37.1	external_calibration_result	internal traversal cost and carrier thresholds can be translated into plasma/energy hypotheses only with unit conversion and measured observables
Proposition 37.2	normal_form_result	energy-plasma translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 37.3	falsifier_or_open_obligation	laboratory plasma behavior, joule conversion, and universal energy bounds require external data

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 37.1	external target or comparator	internal traversal cost and carrier thresholds can be translated into plasma/energy hypotheses only with unit conversion and measured observables	dataset, measured value, uncertainty, comparator, calibration protocol, or external benchmark	only the calibrated target, scale, units, and comparator stated in the row	attach dataset/citation, uncertainty, pass/fail criterion, and falsifier
Proposition 37.2	translation grammar	energy-plasma translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 37.3	obligation/falsifier	laboratory plasma behavior, joule conversion, and universal energy bounds require external data	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-37 defines or uses Plasma, Energy, And Traversal Calibration as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-38 - Observer, Computation, And AI Runtime Translation

Claim	Lane	Statement
Theorem 38.1	cam_crystal_reapplication_result	observer-face selection, delay buffers, and corpus ribbons can be translated into AI/runtime protocols while preserving local receipt state
Proposition 38.2	normal_form_result	observer-runtime translation can be imported by later papers only through its declared source, receipt, and lane.
Protocol 38.3	falsifier_or_open_obligation	consciousness, human perception, and physical collapse claims remain empirical or philosophical unless separately evidenced

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 38.1	translation grammar	observer-face selection, delay buffers, and corpus ribbons can be translated into AI/runtime protocols while preserving local receipt state	CAM/crystal route, source node, projection, window, or addressable evidence artifact	routing and reapplication evidence; not a standalone physical proof	check the CAM source, route, and attached data are named
Proposition 38.2	translation grammar	observer-runtime translation can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 38.3	obligation/falsifier	consciousness, human perception, and physical collapse claims remain empirical or philosophical unless separately evidenced	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-38 defines or uses Observer, Computation, And AI Runtime Translation as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-39 - Falsifiers, Comparators, And Standards Of Evidence

Claim	Lane	Statement
Theorem 39.1	normal_form_result	every FLCR claim can be evaluated by lane, source, receipt, comparator, and falsifier before final admission
Proposition 39.2	normal_form_result	evidence comparator can be imported by later papers only through its declared source, receipt, and lane.
Protocol 39.3	falsifier_or_open_obligation	claims that lack comparator or falsifier fields remain obligations regardless of rhetorical strength

Reviewer Claim Ledger

This ledger expands the formal-claims table into review actions. It is intended to make proof granularity explicit: what is being claimed, what evidence type can support it, what boundary remains, and what the next review action is.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 39.1	translation grammar	every FLCR claim can be evaluated by lane, source, receipt, comparator, and falsifier before final admission	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Proposition 39.2	translation grammar	evidence comparator can be imported by later papers only through its declared source, receipt, and lane	definitions, normal-form construction, conversion rule, and downstream-use boundary	internal formal coherence; no measured physical identity without a separate calibration row	check that the normal form is named and the conversion rule is explicit
Protocol 39.3	obligation/falsifier	claims that lack comparator or falsifier fields remain obligations regardless of rhetorical strength	explicit missing derivation, adapter, receipt, dataset, comparator, or failed test condition	does not negate satisfied lower-level rows	name the next binding action and owner surface

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR - 39 defines or uses Falsifiers, Comparators, And Standards Of Evidence as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.
Imported theorem or external definition	The Standard Model or adjacent physics object is closed only as an external target definition when the relevant definition/citation row is present.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Physical or calibration-facing claim	Measured values, physical identity, and predictive accuracy require scale, units, dataset, uncertainty, comparator, and pass/fail criteria.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

FLCR-40 - Grand Reconstruction Map And Trust-Removal Protocol

Claim	Lane	Statement
Theorem 40.1	grand_synthesis_claim	The strongest integrated FLCR claims are reviewable only when FLCR-40 provides a route from each claim segment to prior proof, analog reconstruction, code/tool solver, comparator, calibration, or named blocker.
Proposition 40.2	normal_form_result	The reconstruction map can be imported by later papers only through declared source routes, trust-removal steps, and claim-lane boundaries.
Protocol 40.3	falsifier_or_open_obligation	A grand-claim segment without a reconstruction route remains blocked, even if the surrounding map is coherent.

Reviewer Claim Ledger

This ledger treats FLCR-40 as a map paper. It is intended to make the review burden explicit: the paper is judged by route completeness, not by whether it reprints every prior proof.

Formal row	Type	Claim in review terms	Evidence required	Boundary	Next review action
Theorem 40.1	reconstruction-map synthesis	Each grand-claim segment must have a route to prior proof, analog reconstruction, code/tool solver, comparator, calibration, or named blocker.	dependency map, prior-proof index, analog route, code/tool route, blocker table	route completeness only; underlying proof remains owned by the routed source	build the prior-proof/reconstruction index
Proposition 40.2	trust-removal protocol	Later papers may import FLCR-40 only as a map and reconstruction protocol.	declared route, trust-removal steps, source paper, evidence lane	does not promote a routed claim beyond its source lane	check that every import names the route and lane
Protocol 40.3	obligation/falsifier	A claim segment with no reconstruction route remains blocked.	missing route, missing solver, missing analog exercise, missing comparator, or missing receipt	does not negate routed lower-level rows	add the missing route or mark the blocker

Claim Granularity And Review Table

The table below separates the claim types so the paper is reviewable without accepting the whole FLCR vocabulary at once.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Formal-system result	FLCR-40 defines or uses Grand Reconstruction Map And Trust-Removal Protocol as a typed FLCR object with declared inputs, operations, outputs, and residue.	Definitions, formal claims, construction steps, and downstream-use rules are internally consistent and lane typed.	External physical truth, measured prediction, or identity with a standard theory.
Computational or receipt-bound result	Enumerations, TarPit runs, CAM/crystal routes, verifier rows, and generated artifacts are claims about the stated finite or executable domain.	A named receipt, validator, manifest, pass report, or rebuild result exists and matches the row scope.	Completeness outside the enumerated domain or physical correctness outside the tested comparator.

Claim type	What this paper may claim	Acceptance test	What is not claimed by that row
Imported theorem or external definition	External objects are inherited from cited papers and definition layers; the synthesis paper does not silently upgrade their scope.	The source theorem, definition layer, or citation target is named and the mapping into this paper is explicit.	A new proof of the imported theorem or a hidden change in the imported object's meaning.
Interpretive bridge or analogy	Analog, workbook, visual, or translation language may explain why the construction is useful.	The analogy preserves the relevant structure and does not promote itself into a theorem.	That the analogy alone proves the formal, computational, or physical claim.
Physical or calibration-facing claim	Global physical synthesis is stated only where its dependency rows supply calibration, comparator, or falsifier coverage.	A dataset, citation, calibration protocol, uncertainty, comparator, or falsifier is attached.	A physical conclusion supported only by shared vocabulary rather than calibration, comparator data, or a stated falsifier.
Open obligation or falsifier	A missing derivation, adapter, receipt, dataset, or failed comparator is a named research channel.	The next binding action and the reason the stronger claim is not closed are stated.	That the base formal result is false merely because a stronger downstream claim remains unfinished.

Appendix C. Key Source Paths

ID	Formal	Companion	Workbook	PDF	Assembled product
FLCR-01	FLCR-01/formal.md	FLCR-01/companion.md	FLCR-01/workbook.md	FLCR-01/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-01_Grounding_Contract_And_Axiom_Discipline__assembled.md
FLCR-02	FLCR-02/formal.md	FLCR-02/companion.md	FLCR-02/workbook.md	FLCR-02/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-02_The_LCR_Carrier__assembled.md
FLCR-03	FLCR-03/formal.md	FLCR-03/companion.md	FLCR-03/workbook.md	FLCR-03/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-03_Correction_Surface_And_Residual_Accounting__assembled.md
FLCR-04	FLCR-04/formal.md	FLCR-04/companion.md	FLCR-04/workbook.md	FLCR-04/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-04_D4_J3_0_Triality_And_Representation_Transport__assembled.md
FLCR-05	FLCR-05/formal.md	FLCR-05/companion.md	FLCR-05/workbook.md	FLCR-05/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-05_Typed_Boundary_Repair__assembled.md
FLCR-06	FLCR-06/formal.md	FLCR-06/companion.md	FLCR-06/workbook.md	FLCR-06/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-06_Oloid_Path_Carrier_And_Transport_Geometry__assembled.md
FLCR-07	FLCR-07/formal.md	FLCR-07/companion.md	FLCR-07/workbook.md	FLCR-07/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-07_Causal_Links_And_Obligation_Ledgers__assembled.md
FLCR-08	FLCR-08/formal.md	FLCR-08/companion.md	FLCR-08/workbook.md	FLCR-08/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-08_Discrete_Continuous_Bridge_Without_Physical_Overclaim__assembled.md
FLCR-09	FLCR-09/formal.md	FLCR-09/companion.md	FLCR-09/workbook.md	FLCR-09/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-09_Lattice_Closure_And_Terminal_Addresses__assembled.md
FLCR-10	FLCR-10/formal.md	FLCR-10/companion.md	FLCR-10/workbook.md	FLCR-10/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-10_Temporal_Windows_And_Hamiltonian_Readouts__assembled.md
FLCR-11	FLCR-11/formal.md	FLCR-11/companion.md	FLCR-11/workbook.md	FLCR-11/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-11_Master_Receipt_And_Stack_Replay__assembled.md
FLCR-12	FLCR-12/formal.md	FLCR-12/companion.md	FLCR-12/workbook.md	FLCR-12/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-12_The_ory_Admission_Gates__assembled.md
FLCR-13	FLCR-13/formal.md	FLCR-13/companion.md	FLCR-13/workbook.md	FLCR-13/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-13_Cellular_Automata_Prediction_Surfaces__assembled.md

ID	Formal	Companion	Workbook	PDF	Assembled product
FLCR-14	FLCR-14/formal.md	FLCR-14/companion.md	FLCR-14/workbook.md	FLCR-14/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-14__Quark_Face_Algebra_Before_Standard_Mode l_Translation__assembled.md
FLCR-15	FLCR-15/formal.md	FLCR-15/companion.md	FLCR-15/workbook.md	FLCR-15/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-15__Curvature_As_Boundary_Repair_Demand__assembled.md
FLCR-16	FLCR-16/formal.md	FLCR-16/companion.md	FLCR-16/workbook.md	FLCR-16/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-16__Mass_Residue_And_Carrier_Accounting__assembled.md
FLCR-17	FLCR-17/formal.md	FLCR-17/companion.md	FLCR-17/workbook.md	FLCR-17/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-17__Continuum_Edge_Residuals__assembled.md
FLCR-18	FLCR-18/formal.md	FLCR-18/companion.md	FLCR-18/workbook.md	FLCR-18/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-18__Exceptional_Towers_V0A_Routes_And_Observer_Face_Selection__assembled.md
FLCR-19	FLCR-19/formal.md	FLCR-19/companion.md	FLCR-19/workbook.md	FLCR-19/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-19__Layer_2_Synthesis_Ledger__assembled.md
FLCR-20	FLCR-20/formal.md	FLCR-20/companion.md	FLCR-20/workbook.md	FLCR-20/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-20__Applied_Forge_Reader_And_Descriptor_Kernel__assembled.md
FLCR-21	FLCR-21/formal.md	FLCR-21/companion.md	FLCR-21/workbook.md	FLCR-21/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-21__Materials_Candidate_Generation__assembled.md
FLCR-22	FLCR-22/formal.md	FLCR-22/companion.md	FLCR-22/workbook.md	FLCR-22/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-22__Protein_Descriptor_And_Fold_Facing_Kernel__assembled.md
FLCR-23	FLCR-23/formal.md	FLCR-23/companion.md	FLCR-23/workbook.md	FLCR-23/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-23__Finite_Game_Lattices_And_Local_Rule_Automata__assembled.md
FLCR-24	FLCR-24/formal.md	FLCR-24/companion.md	FLCR-24/workbook.md	FLCR-24/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-24__Energetic_Traversal_Maps__assembled.md
FLCR-25	FLCR-25/formal.md	FLCR-25/companion.md	FLCR-25/workbook.md	FLCR-25/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-25__Shear_Plasma_And_Carrier_Horizons__assembled.md
FLCR-26	FLCR-26/formal.md	FLCR-26/companion.md	FLCR-26/workbook.md	FLCR-26/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-26__Observer_Delay_And_Shared_State_Protocols__assembled.md

ID	Formal	Companion	Workbook	PDF	Assembled product
FLCR-27	FLCR-27/formal.md	FLCR-27/companion.md	FLCR-27/workbook.md	FLCR-27/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-27__Mon ster_Ceiling_And_L arge_Invariant_Bou ndaries__assembled .md
FLCR-28	FLCR-28/formal.md	FLCR-28/companion.md	FLCR-28/workbook.md	FLCR-28/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-28__CAM _Crystal_Projector s_And_MannyAI_Runt ime__assembled.md
FLCR-29	FLCR-29/formal.md	FLCR-29/companion.md	FLCR-29/workbook.md	FLCR-29/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-29__Cor pus_Ribbon_And_Ret rospective_LCR_Rea dout__assembled.md
FLCR-30	FLCR-30/formal.md	FLCR-30/companion.md	FLCR-30/workbook.md	FLCR-30/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-30__Sup ervisor_Cursor_And _Universal_Normal_ Form_Intake__assem bled.md
FLCR-31	FLCR-31/formal.md	FLCR-31/companion.md	FLCR-31/workbook.md	FLCR-31/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-31__Gau ge_Groups_Translat ed_Into_LCR__assem bled.md
FLCR-32	FLCR-32/formal.md	FLCR-32/companion.md	FLCR-32/workbook.md	FLCR-32/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-32__QCD _And_Color_Transpo rt_In_LCR__assembl ed.md
FLCR-33	FLCR-33/formal.md	FLCR-33/companion.md	FLCR-33/workbook.md	FLCR-33/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-33__Ele ctroweak_Higgs_And _Mass_Residue Tran slation__assembled .md
FLCR-34	FLCR-34/formal.md	FLCR-34/companion.md	FLCR-34/workbook.md	FLCR-34/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-34__Ele ctron_Hole_Exciton _Accounting__assem bled.md
FLCR-35	FLCR-35/formal.md	FLCR-35/companion.md	FLCR-35/workbook.md	FLCR-35/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-35__GR_ Curvature_And_Cont inuum_Translation_ __assembled.md
FLCR-36	FLCR-36/formal.md	FLCR-36/companion.md	FLCR-36/workbook.md	FLCR-36/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-36__Con densed_Matter_Mate rials_And_Metamate rials__assembled.m d
FLCR-37	FLCR-37/formal.md	FLCR-37/companion.md	FLCR-37/workbook.md	FLCR-37/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-37__Pla sma_Energy_And_Tra versal_Calibration __assembled.md
FLCR-38	FLCR-38/formal.md	FLCR-38/companion.md	FLCR-38/workbook.md	FLCR-38/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-38__Obs erver_Computation_ And_AI_Runtime_Tra nslation__assemble d.md
FLCR-39	FLCR-39/formal.md	FLCR-39/companion.md	FLCR-39/workbook.md	FLCR-39/paper.pdf	ASSEMBLED_FLCR_PRO DUCTS/FLCR-39__Fal sifiers_Comparator s_And_Standards_Of _Evidence__assembl ed.md

ID	Formal	Companion	Workbook	PDF	Assembled product
FLCR-40	FLCR-40/formal.md	FLCR-40/companion.md	FLCR-40/workbook.md	FLCR-40/paper.pdf	ASSEMBLED_FLCR_PRODUCTS/FLCR-40__Grand_Reconstruction_Map_And_Trust_Removal_Protocol__assembled.md

Appendix D. Assembled FLCR Products (14-section unified form)

Each slot below has a curated assembled markdown product with honest assembly status, blockers, and evidence intake. Status reflects pipeline gates — not publication inflation.

ID	Assembly status	Blockers	Evidence items	Assembled path
FLCR-01	ASSEMBLE	0	2425	ASSEMBLED_FLCR_PRODUCT S/FLCR-01__Grounding_Contract_And_Axiom_Discipline__assembled.md
FLCR-02	ASSEMBLE	0	982	ASSEMBLED_FLCR_PRODUCT S/FLCR-02__The_LCR_Carrier__assembled.md
FLCR-03	ASSEMBLE	0	778	ASSEMBLED_FLCR_PRODUCT S/FLCR-03__Correction_Surface_And_Residual_Accounting__assembled.md
FLCR-04	ASSEMBLE	0	866	ASSEMBLED_FLCR_PRODUCT S/FLCR-04__D4_J3_0_Triality_And_Representation_Transport__assembled.md
FLCR-05	ASSEMBLE	0	877	ASSEMBLED_FLCR_PRODUCT S/FLCR-05__Typed_Boundary_Repair__assembled.md
FLCR-06	ASSEMBLE	0	1642	ASSEMBLED_FLCR_PRODUCT S/FLCR-06__Oloid_Path_Carrier_And_Transport_Geometry__assembled.md
FLCR-07	ASSEMBLE	0	1083	ASSEMBLED_FLCR_PRODUCT S/FLCR-07__Causal_Links_And_Obligation_Ledgers__assembled.md
FLCR-08	ASSEMBLE	0	4635	ASSEMBLED_FLCR_PRODUCT S/FLCR-08__Discrete_Continuous_Bridge_Without_Physical_Overclaim__assembled.md
FLCR-09	ASSEMBLE	0	3644	ASSEMBLED_FLCR_PRODUCT S/FLCR-09__Lattice_Closure_And_Terminal_Addresses__assembled.md
FLCR-10	ASSEMBLE	0	472	ASSEMBLED_FLCR_PRODUCT S/FLCR-10__Temporal_Windows_And_Hamiltonian_Readouts__assembled.md
FLCR-11	ASSEMBLE	0	1091	ASSEMBLED_FLCR_PRODUCT S/FLCR-11__Master_Receipt_And_Stack_Replay__assembled.md
FLCR-12	ASSEMBLE	0	2039	ASSEMBLED_FLCR_PRODUCT S/FLCR-12__Theory_Admission_Gates__assembled.md
FLCR-13	ASSEMBLE	0	938	ASSEMBLED_FLCR_PRODUCT S/FLCR-13__Cellular_Automata_Prediction_Surfaces__assembled.md

ID	Assembly status	Blockers	Evidence items	Assembled path
FLCR-14	ASSEMBLE	0	234	ASSEMBLED_FLCR_PRODUCT S/FLCR-14__Quark_Face_ Algebra_Before_Standar d_Model_Translation__a ssembled.md
FLCR-15	ASSEMBLE	0	219	ASSEMBLED_FLCR_PRODUCT S/FLCR-15__Curvature_A s_Boundary_Repair_Dema nd__assembled.md
FLCR-16	ASSEMBLE	0	75	ASSEMBLED_FLCR_PRODUCT S/FLCR-16__Mass_Residu e_And_Carrier_Accounti ng__assembled.md
FLCR-17	ASSEMBLE	0	4951	ASSEMBLED_FLCR_PRODUCT S/FLCR-17__Continuum_E dge_Residuals__assembl ed.md
FLCR-18	ASSEMBLE	0	1214	ASSEMBLED_FLCR_PRODUCT S/FLCR-18__Exceptional _Towers_VOA_Routes_And _Observer_Face_Selecti on__assembled.md
FLCR-19	ASSEMBLE	0	332	ASSEMBLED_FLCR_PRODUCT S/FLCR-19__Layer_2_Syn thesis_Ledger__assembl ed.md
FLCR-20	REROUTE	2	984	ASSEMBLED_FLCR_PRODUCT S/FLCR-20__Applied_For ge_Reader_And_Descript or_Kernel__assembled.m d
FLCR-21	ASSEMBLE	0	4882	ASSEMBLED_FLCR_PRODUCT S/FLCR-21__Materials_C andidate_Generation__a ssembled.md
FLCR-22	ASSEMBLE	0	3040	ASSEMBLED_FLCR_PRODUCT S/FLCR-22__Protein_Des criptor_And_Fold_Facin g_Kernel__assembled.md
FLCR-23	ASSEMBLE	0	924	ASSEMBLED_FLCR_PRODUCT S/FLCR-23__Finite_Game _Lattices_And_Local_Ru le_Automata__assembled .md
FLCR-24	DISCOVERY	4	2943	ASSEMBLED_FLCR_PRODUCT S/FLCR-24__Energetic_T raversal_Maps__assembl ed.md
FLCR-25	ASSEMBLE	0	876	ASSEMBLED_FLCR_PRODUCT S/FLCR-25__Shear_Plasm a_And_Carrier_Horizons __assembled.md
FLCR-26	ASSEMBLE	0	97	ASSEMBLED_FLCR_PRODUCT S/FLCR-26__Observer_De lay_And_Shared_State_P rotocols__assembled.md
FLCR-27	ASSEMBLE	0	387	ASSEMBLED_FLCR_PRODUCT S/FLCR-27__Monster_Cei ling_And_Large_Invaria nt_Boundaries__assembl ed.md
FLCR-28	ASSEMBLE	0	3217	ASSEMBLED_FLCR_PRODUCT S/FLCR-28__CAM_Crystal _Projectors_And_MannyA I_Runtime__assembled.m d

ID	Assembly status	Blockers	Evidence items	Assembled path
FLCR-29	ASSEMBLE	0	4828	ASSEMBLED_FLCR_PRODUCT S/FLCR-29__Corpus_Ribbon_And_Retrospective_LCR_Readout__assembled.md
FLCR-30	ASSEMBLE	0	422	ASSEMBLED_FLCR_PRODUCT S/FLCR-30__Supervisor_Cursor_And_Universal_Normal_Form_Intake__assembled.md
FLCR-31	ASSEMBLE	0	781	ASSEMBLED_FLCR_PRODUCT S/FLCR-31__Gauge_Groups_Translated_Into_LCR__assembled.md
FLCR-32	ASSEMBLE	0	751	ASSEMBLED_FLCR_PRODUCT S/FLCR-32__QCD_And_Color_Transport_In_LCR__assembled.md
FLCR-33	ASSEMBLE	0	758	ASSEMBLED_FLCR_PRODUCT S/FLCR-33__Electroweak_Higgs_And_Mass_Residue_Translation__assembled.md
FLCR-34	ASSEMBLE	0	740	ASSEMBLED_FLCR_PRODUCT S/FLCR-34__Electron_Hole_Exciton_Accounting__assembled.md
FLCR-35	ASSEMBLE	0	832	ASSEMBLED_FLCR_PRODUCT S/FLCR-35__GR_Curvature_And_Continuum_Translation__assembled.md
FLCR-36	ASSEMBLE	0	1566	ASSEMBLED_FLCR_PRODUCT S/FLCR-36__Condensed_Matter_Materials_And_Metamaterials__assembled.md
FLCR-37	ASSEMBLE	0	773	ASSEMBLED_FLCR_PRODUCT S/FLCR-37__Plasma_Energy_And_Traversal_Calibration__assembled.md
FLCR-38	ASSEMBLE	0	2034	ASSEMBLED_FLCR_PRODUCT S/FLCR-38__Observer_Computation_And_AI_Runtime_Translation__assembled.md
FLCR-39	REROUTE	2	1661	ASSEMBLED_FLCR_PRODUCT S/FLCR-39__Falsifiers_Comparators_And_Standards_Of_Evidence__assembled.md
FLCR-40	ASSEMBLE	0	2067	ASSEMBLED_FLCR_PRODUCT S/FLCR-40__Grand_Reconstruction_Map_And_Trust_Removal_Protocol__assembled.md